

A Packing of 76 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 76 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.898754. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.897902, obtained by deleting 1 of the 77 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 76 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.898754\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 76$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 0.897902, obtained by deleting 1 of the 77 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 76/75 = 0.935385$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N76.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 75 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
898754737362121591914498827709624470739033210224
denominator:
1000000000000834975327086150220920575220612995025

B Coordinates

76 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N76.txt`.

100411816632	486746101484	-521095835161	-97578872141
-511826396579	-247606265070	-15054331700	61582403057
246506026566	-183751089750	285888243452	-69364882346
212822339532	246251142239	-249734049280	140702434852
204352116614	14620763956	-618463471907	-198118385480
297513459700	346609845902	-101133886442	-518628620557
403644403585	184131350328	97235619501	-101869747205
-447840881194	8577169903	-73008075818	398770200528
221573129128	229250741759	-69085564335	-116169475647
116940336078	163542497514	6644331110	-166632178457
-29097301913	259407005190	-146016948143	614659147922
-161501415494	-167826264225	70761103498	227867065979
-196033869236	-185101003110	-171258553911	-149060767436
-336528373110	496611902641	29490195610	126820581757
-449191589625	537637875829	-533113249972	172446547291
-121033305177	-89645632330	429614809168	618701896569
69987982329	64386297279	355713677388	43897448668
-245920033724	261873057134	-402202691738	-17184849635
101357181666	-300146537025	-273955747086	-1567912287
-229856290351	112703836076	-296700475787	-461056866095
-59804374304	267862874995	-133148860036	-173281692308
-1108132438	243847957750	291436148946	132150968801
-274528431627	-16864705124	-249629316088	87823216871
-447461610490	-259898863811	56457382873	-476699739206
-13185809077	460356316038	-540158564842	231352891141
-512352346004	-140454354552	-231077702027	-274704640099
113234458237	614972553270	-66672457566	21094940870
219769433985	183587664708	160731334973	80220983447
-351953388607	104617548202	-70739050213	289368347258
174375463687	-148682677687	118914896106	176911265693
77815132366	74171828578	591071655613	-618059559684
267418052513	-144490830457	-484538971556	-398346807301
-178649714490	211419847366	9128818471	105683274404
156512899629	-353590891198	-226865677818	-374283530559
-34537509534	-404695361789	-16133136499	-154101873719
-765530104173	122542249048	227182096773	-178270264101
225596224527	359334165995	64258187255	40479290278
106609226256	-170539252306	405521192681	-86387693763
-33665736143	-43745154475	-53134306754	181405840441
-36210996966	53425399143	139576329634	144579942939
-388568249921	-320285063726	-177444587084	172600625822
3164673509	41156709650	-275543528315	-839950573816
455208367037	-346509398258	-324575324334	65513128722
183789522869	-573509632146	-126173980105	-141663726939
-465174334404	-7661424429	150986426929	209662724098
-75498145370	-314972048783	525500713423	-72099359182
-334957843911	-75550798298	-199359727645	-174836059527
188845521106	-244642104738	-17341286653	121473065355
-58093704723	19716526410	343637147076	13072981339
-260263278751	-432933673152	-255965381422	-290527402904
403510554501	169519386216	-234294723166	160393243254
-53113819175	261958490469	-484956936667	-125218406657
-170776893803	-24583451892	167178185308	468127431377
-332199110102	-49376022938	126614282201	44858732159
-565609583741	-279986699023	148581049469	322889282246
-223769161513	-554430171696	67317855444	-240957091034
-7569677994	102317451031	-336438621167	658157908686
385763936501	160308093768	-134266198890	134925559331
227619550750	386579658335	158714554748	-27401018902
-138998613936	283781215340	218499034006	-154735253958
-66607754955	-458282183315	465298123743	-30319348375
345736695605	6827543267	-141045776785	299364075163
-343922000010	305937447244	394200792206	92262048644
-167188127223	-67500377513	135002206687	315267867003
16885140024	144511165982	354389208989	-388349779985
592193410199	135080853847	220764817773	151647412874
158488576651	58813493075	-109580974697	-321888617409
-109003460320	-494364576430	441548286019	350078492778
-182794793207	194202229083	253329108824	-50807854124
-178683668311	-116520517949	-34821183887	-170079291404

396659039413	-118303785565	193520477956	88101210949
157976220167	-334053891171	149553892112	156695415482
-398030873382	327761221458	118935500871	-382888923490
344998621565	-396131499020	282375274131	251248233857
-29179558926	3135915948	-450351438371	-277402968374
92298941861	-145069479015	-392370067481	265567604591
116766805715	-481718933347	-282797184710	-187704198528
-369606858577	385723948252	-534715113714	365153093548
-2181331204	-9205331448	-88002160846	146351016687
53470378464	206717429567	66718877005	448227044962
99651666519	-116610942001	-193212963245	-13083780730
4598359446	-360575386355	360808651038	-422867961017
477859659860	243091958980	-349471558132	-64247646751
111271458490	469980385665	291816959887	-142109076478
-599388984636	-6826759471	-7083902377	-146559617574
289746569329	-277335315044	-340089592702	-520165314384
196603362509	79176560185	-177410414477	14646914946
-316622496715	-283276140927	-340917646628	13720952000
118687552569	-90237130952	75128721388	185505839444
269864171909	-191889129013	-344760524144	233177651620
-431132585462	-570223835668	-123621535398	-262903476602
-191041710998	254998240141	-214161965157	775482017
337833472148	-95236498755	419685805509	-107487586423
119619168406	357132452474	-154644290604	127717355669
-105638971236	220655073060	64429443348	705304473701
-390570255026	-198119780057	-554066634010	14082849472
-5483489598	26297240906	-29821754275	-232177200211
-215632196862	171834951496	165199966527	173568744035
513191506141	-405270714248	-388411332711	-73015071638
-137072757842	-307873491872	41528463388	-281005422917
-186730915854	351481830378	153416738932	-410846795729
-221598645274	346920162551	31865528268	-6354707091
-349099771589	420331917540	432641659307	248451053663
39542375916	24514858123	198700591906	67420260008
-177867639524	-82795223202	-437098312979	-624996473827
-68765520229	-306593819073	204714530981	-363278808935
82046596436	55371383427	184292550103	146060198539
150203955657	-62708378868	-181363270901	228996741131
324532133925	-24659319964	510410580359	60431054086
157174421286	488489475521	-77971890618	276401183742
-131704198767	445580936223	164511670863	466038695490
-116581101756	334649850633	227477023	29043733041
-512975377302	-127065269052	-120658816259	-359258511638
620321379853	-491122118110	-182821965163	-289067271375
-44753522797	-193693398452	578647393291	364756751776
-216682933206	-216019987640	181835360341	9725005729
505088523827	29251018245	104336311310	-240740824085
143270354492	-254015615556	327612603401	-109037615376
-184593646008	373639933288	324631319184	-71867599740
412345396178	-81033827097	-515969799205	-438186033817
80985164154	-44517965345	-108707204869	298544706645
426596072695	-172755060645	-28231345947	-230762195806
-149071419771	-139944875849	-44457174415	75638776050
-487625941910	41025964786	-235148155340	-273894906745
63064763171	-203525694226	132915600084	-174787640622
-117746251231	-255801418863	260031945801	282528982490
-79871363835	-95421825041	-30842636343	232534594603
-23221173982	778757867485	173488639356	-87464696534
-63280308103	-50023533205	148730930633	-223404277581
143886201160	-106209034704	217928967583	533287155990
237560475890	90323679201	-552290044802	-141930734725
519550040996	-326674128848	-415835113593	480526069049
436265388007	-293953436516	-436674404049	-236153049276
122701024032	112900929943	-319203141780	229762589106
58255258197	-29057824410	411302578925	562932671743
139259979954	391130324249	360294310046	258317888138
-140709614171	157185132770	-285437158149	483339082243
480526656287	-243164637834	297786868871	63215222606
-240946065123	300020778355	107990548879	328779785349
309952805746	-302170801118	255555857706	-176550490730
126191444686	-103641037614	324505277659	-3506891543
-196413005986	356183363987	134633659267	-260340579209
-297050088108	478935481021	-142507048051	26666352146
116992179208	651440246956	115652924522	172365336908
-559832257873	-139276382281	64862293627	-308683063806
-127783362390	-204495494920	142808942897	-176644445971
543897033605	-8623978831	370125246202	-121658154601
-334503031091	32541380663	-179662314088	-380536382451
77247296178	-134969227756	-225540939403	-361982466530
283474182118	107482997391	114836287652	-40960603517
-82796964826	-4124599637	-533944732771	-352844118204
-116022868021	246883130671	294009135572	330306186354
33108779285	-349593922394	111263119752	-114952107890
119588500109	-185535800604	200643799215	79583554511
581878668790	219769700677	88882385932	-327939961909
-328405070834	-466476395071	-514471946086	-63528054373
293789899967	-100730642162	73125778487	-490802098301
556216307706	2237958851	210175321551	13098866942
-28630896859	-112768901590	-372187501259	288670451120
87861746047	-147842018559	-268038468441	-127978365928
29135521420	-315939648620	379841456637	590627827208
154228660997	247561225176	-120639205395	-81667683338
-118383130524	-277013790885	515273662877	-311341427300
-249283712980	147193427665	-38612534419	-197022430718
-1233342878	-165625380814	-273019404084	49008415802
-151011771271	607553727687	49490095056	104429690194
160438316731	-87407473073	325843484659	281234814217
206517600226	513927421620	106833556132	95291126576
78223602868	165688003629	-86366937572	224611612521
93547421142	272015120734	-172979862454	-72184335111
-12421454090	-103046153240	-363102034376	-93234050000
387942886033	-202075463857	201475337475	-485835200943

326906174386	92993965087	447094724238	65141187739
281247881392	109289395315	119466793091	18435590279
-87546818058	-80664544048	243173974146	508800483533
318814900489	307563873564	-61819162463	-101055889688
-361719366089	-204059291808	595690485875	5588517879
123553000157	260121385467	-69469488259	-305909499198
287323028832	621265318877	241784568310	288220645241
170456314889	-329847856682	-16350591036	-420233319829
-431990437929	64947440027	104554957569	-426974123520
-286284101629	-13725262311	253273795986	-225021326173
-320443154830	-66509644627	102439970167	472946454397
256044309744	437758417475	-959743843	31003216442
-202287903647	-5280946053	271483530705	-54955744419
275932774337	109375569853	79934725768	-166759986660
-114678467173	-372290538167	470493251739	-415176940778
-59112621490	142701361567	-575287538693	77357924324
125456320054	156553890049	-150239479508	190755469226
-127676178519	57800815842	292892679002	-287019749657
-424141689362	169184581576	376344295794	-23712228135
-27738479849	45090624382	178193330863	269041086074
-462362146810	321376607459	-133396825873	-13333443339
-330343138755	-343331560182	-371509588029	-317591681147
-117362311867	524363870485	-6851137676	-466642244969
-279117552015	51891207412	-209802319976	-245687973348
247970107512	-309117956905	-390119305210	105812709502
339191274369	237500868383	-70161436978	9613729562
423701188790	-256684562050	-164113748411	481115621895
103237186445	-22611657220	622812944375	-60745541249
-424580268002	-479320438452	-144508768345	-49825141800
-559291683902	138151802752	254417544067	50257109571
272078496879	62454941828	-167473046732	584952633020
292600530885	235764415586	-337313202347	5575426221
-183189456887	454561476780	27175116259	-201357219800
154143390956	173927867013	-91820182671	323521324333
417475426269	413613904610	243114067731	-427419143030
-135898578581	534523263807	207432327724	134460079153
-213213467338	166487795084	-390151312504	-144837885438
-127512292858	212056627332	495699423812	295986538009
83651605683	259783240535	187286748250	-160312696338
133547881417	-114149762374	-509136196951	603508418994
272129188609	281148215182	-62601776602	-205520518123
134460281730	-83351295069	-196563231337	231436027865
-101055881614	-87466811654	46365558243	-167087580834
-15012789265	7826720923	302628770427	-36479612496
-158353551515	526655150707	223533566545	239584548531
187700228657	180600449806	104628846922	118523587583
146261462609	448032075074	-377618248418	370491423807
692739372979	25144573809	-209014888167	327859737024
388145069528	46969577167	194188867532	245431344189
-129036454738	133738507576	-391909363455	-227096969708
-64073967992	227277528019	-471337713080	-488552468249
367006132478	55590378382	154841455939	-106589059513
172438419610	-363889917492	273047052273	-362560531891
-16871479441	-512779453251	-226883234492	42974465878
-545591108283	-47720273263	-477776951355	-204560886197
432951154023	-126957566171	29956572537	396956786760
45235925289	-144552063170	-70964792756	266022647108
-291143621219	-16790014093	103954045714	530169211835
-456938471121	407463407847	-72377493300	67039839912
81619676388	303164784172	-319972680388	-65673020288
-56457581989	-270856835790	-321420968314	64730335782
-64411780911	-338322074880	-50352312967	28649894569
-82409246075	-534514195570	358804132010	442870909923
235671869227	-70025249748	338382908123	41627193191
-97488569313	115563967351	-131324482271	271310653050
-364911890574	78552406386	279470386884	-511110286099
-140889545685	-369777696610	95157365431	-472120384921
-13634390591	-288262704950	88303781156	-338444450489
-188812069937	55649129162	200257594787	15431457593
-156039482080	154340093892	581134030972	138017079452
-56735896742	-261929128569	199410314645	21747827026
-234145991564	144018691425	-136896505780	149929803169
664270970645	10521128088	-123793469035	113710722547
-544660394644	-400765949027	-18227537981	250381889787
48139296296	175649376978	352773195712	109485423110